In [1]:
```
import numpy as np
import pandas as pd
```

In [3]:
```
sales ={"product_id":[1,2,3,4,5],
       "product_name":["laptop","mobile","mouse","tv","keyboard"],
       "quantity_sold":[5,3,9,4,7]}
df=pd.DataFrame(sales)
df
```

Out[3]:

	product_id	product_name	quantity_sold
0	1	laptop	5
1	2	mobile	3
2	3	mouse	9
3	4	tv	4
4	5	keyboard	7

In [4]:
```
inventory = {"product_id":[1,2,3,4,5],
             "stocks_available":[20,12,16,13,11]}
df1=pd.DataFrame(inventory)
df1
```

Out[4]:

	product_id	stocks_available
0	1	20
1	2	12
2	3	16
3	4	13
4	5	11

Inner Merge

In [7]:
```
IM=pd.merge(df,df1,on="product_id",how="inner")
IM
```

Out[7]:

	product_id	product_name	quantity_sold	stocks_available
0	1	laptop	5	20
1	2	mobile	3	12
2	3	mouse	9	16
3	4	tv	4	13
4	5	keyboard	7	11

left merge

In [8]:
```
LM=pd.merge(df,df1,on="product_id",how="left")
LM
```

Out[8]:

	product_id	product_name	quantity_sold	stocks_available
0	1	laptop	5	20
1	2	mobile	3	12
2	3	mouse	9	16
3	4	tv	4	13
4	5	keyboard	7	11

set index

In [13]:
```
sales_index=df.set_index("product_id")
inv_index=df1.set_index("product_id")
print(sales_index)
print(inv_index)
```

```
           product_name  quantity_sold
product_id
1                laptop              5
2                mobile              3
3                 mouse              9
4                    tv              4
5              keyboard              7
            stocks_available
product_id
1                         20
2                         12
3                         16
4                         13
5                         11
```

In [ ]:
```
RM=sales_index.join
```